

Week 5 Weekly Report

Week 5 designed two machine-learning approaches on top of the existing JPM chooser model. The first predicts forward volatility and sends it to BSM. The second predicts the chooser proxy price directly, as required by the project brief.

1 Work Completed

- Created a 20-day forward realized-volatility target and recorded each label's end date.
- Created the chooser proxy target by passing forward volatility through the Week 4 pricing engine.
- Prepared 17 volatility features and 19 direct-pricing features from the Week 2 data.
- Implemented Approach 1 for volatility prediction followed by BSM pricing.
- Implemented Approach 2 as a direct supervised proxy-price model without the current BSM price as an input.
- Applied chronological 70/15/15 blocks and purged 20 rows at both development boundaries.

2 Deliverables

- Week5_ML_Design_Implementation.ipynb
- ml_features.py
- ml_pipeline.py
- ML_Architecture_Design.md
- ml_config.json
- ml_dataset.csv and split summaries

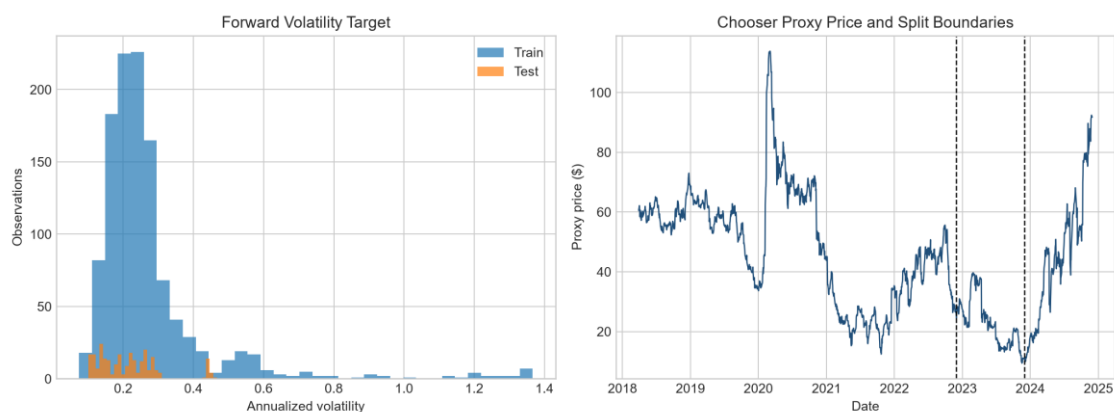
3 Progress Summary

The two model frameworks connect directly to the Week 2 features and Week 4 BSM baseline. The dataset contains 1,680 usable observations. Label-end assertions confirm that training targets stop before validation begins and validation targets stop before testing begins.

4 Experimental Results

Split	Rows	Start	End	Share
Train	1,156	2018-03-29	2022-10-28	68.8%
Purged	20 + 20	2022-10-31	2023-11-29	2.4%
Validation	232	2022-11-29	2023-10-31	13.8%
Test	252	2023-11-30	2024-11-29	15%

Figure 1 Targets and chronological data split



5 Issues Encountered

- A direct price model may fail when JPM price levels in the test period exceed the development range; Week 6 tests this limitation rather than hiding it.
- The target remains a theoretical proxy because actual chooser transactions are unavailable.
- Forward realized volatility must be shifted correctly; otherwise future information would leak into current features.

6 Plan for Week 6

- Tune candidates with expanding-window validation and a 20-day CV gap.
- Select candidates on validation and evaluate the test period once.
- Explain the exact selected models with SHAP and permutation importance.